QUANT

Ratio & Partnership, Mixtures & Alligations

Assignment 05

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Mixture & Partnership Online Assignment -05

1. A mixture consists of 15 parts of coffee, purchased at Rs 2.10 per kg and 1 part of chicory, purchased at 98 paise per kg. If it is sold at Rs 2.25 per kg, what profit would be made on the sale of 5 quintals?

1. Rs. 120 2. Rs. 110 3.Rs. 125 4. Rs. 140

2. Sea water contains 5 % salt by water. How many kg of fresh water must be added to 60 kg of sea water for the content of salt in solution to be made 3 %?

1. 50 kg 2.60 kg 3.45 kg 4. 40 kg

3. A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is:

1. 400 kg 2. 560 kg 3. 600 kg 4. 640 kg

4. There are 15 litres of a 70 % solution of sulphuric acid. How many litres of an 80 % solution of the same acid must be added to obtain 72 % solution of the acid?

1. 19/4 2. 15/4 3.17/3 4. None of these

5. A dishonest milkman fills up his bucket, which is only $^4/_5{}^{th}$ full of milk, with water. He again removes 5 litres of this mixture from the bucket and adds an equal quantity of water. If milk is now 60 % of the mixture, what is the capacity of the bucket in litres?

1.25litres 2.18litres 3.35litres 4.20 litres

6. 300 gm of sugar solution has 40 % sugar in it. How much sugar should be added to make it 50 % in the solution?

1. 55 gm 2.40 gm 3.50 gm 4. 60 gm

7. Vessel A and B contain 4 litres and 6 litres of two different samples of acid. If the contents of two vessels are mixed, the result is 35 % acid concentration. If one litre each from A and B are mixed, the result is 36 % acid concentration. Find the amount of pure acid in A and B.

1.1.64 lt, 1.76 lt 2.1.64 lt, 1.86 lt 3.1.58lt, 1.86 lt 4.1.2lt, 1.8 lt

8. X and Y began a business, with X as the capital provider to the tune of Rs 24lacs and Y as the manager. They have a pact to share the profits in the ratio 3:1 in favour of X upto a profit of Rs 2lacs. All profits in excess of Rs 2Lacs have to be shared equally by X and Y. If the profit is Rs 6lacs, what is Y's share in it?

1. Rs 1,50,000 2. Rs 2,00,000 3. Rs 2,50,000 4. Rs 2,75,000

9. A and B enter into a partnership for a year. A contributes Rs. 1,500 and B Rs. 2,000. After 4 months they admit C, who contributes Rs. 2,250. If B withdraws his contribution after 9 months, how would they share a profit of Rs. 900 at the end of the year?

1. Rs.300 to each person 2. Rs 200, Rs 300, Rs 400
3. Rs 250, Rs 250, Rs 400 4. Rs 100, Rs 300, Rs 500

10. A, B and C enter into a partnership. A advances one-fourth of the capital for one-fourth of the time; B, one-fifth of the capital for one-half of the time; C, the remainder of the capital for all the time. How should they divide a profit of Rs. 3,420?

1. A- 400, B-480, C- 2540
2. A- 500, B-800, C- 2120
3. A- 300, B-480, C- 2640
4. None of these

Answer key & Explanations

Q No.	Key	Explanations
1.	2	CP of the mixture = $\frac{(15\times 2.10)+(1\times 0.98)}{15+1}$ = 2.03/ kg. SP = 2.25/ kg. $\therefore$ Profit per kg = Rs 0.22. Hence profit for 5 quintals i.e. 500 kg = 500 × 0.22 = Rs 110.
2.	4	Let x kg of fresh water be added. Quantity of salt in 60 kg mixture = $60\times\frac{5}{100}$ = 3 kg. $\therefore$ Water in the mixture = 60 – 3 = 57 kg. $\therefore \frac{57+x}{60+x}=\frac{97}{100}\Rightarrow x$ = 40 kg.
3.	3	By the rule of allegation, we have: Profit on 1st part Profit on 2nd part 8% 18% Mean Profit 14% 4 6 Ration of 1st and 2nd parts = 4 : 6 = 2 : 3 $\therefore$ Quantity of 2nd kind = $\left(\frac{3}{5}\times 1000\right)$kg = 600kg.
4.	2	Ratio between the acids of 70 % solution and 80 % solution to obtain 72 % solution is 80 – 72 : 72 – 70 $\Rightarrow$4 : 1. Let x liters of 80 % solution must be added. $\therefore$4 : 1 = 15 : $x\Rightarrow x=\frac{15}{4}=3\frac{3}{4}$ liters.
5.	4	Let the capacity of bucket be x liters. $\frac{4}{5}$th is milk i.e. $\frac{1}{5}$th is water. 5 liters of mixture is removed and replaced by water. Milk is 60 % of mixture i.e. $\frac{3}{5}$. So ratio of milk : water = 3 : 2. $\therefore \frac{\frac{4}{5}(x-5)}{\frac{1}{5}(x-5)+5}=\frac{3}{2}$. Solving this, we get x = 20 liters.
6.	4	Quantity of sugar in the mixture = 40% of 300 = 120g. Let x gm of sugar be added. $\therefore \frac{120+x}{300+x}=\frac{50}{100}\Rightarrow x=60g$.
7.	2	Contents in vessel A and B = 4 liters and 6 liters. Let amount of pure acid in A be x liters and amount of pure acid in B be y liters.

		2 equations can be formed as $\frac{x+y}{4+6}=\frac{35}{100}$ and $\frac{\frac{x}{4}+\frac{y}{6}}{1+1}=\frac{36}{100}\Rightarrow x+y=3.5$ and $\frac{x}{4}+\frac{y}{6}=0.72$. Solving these 2 equations, we get x = 1.64 liters and y = 1.86 liters.
8.	3	Out of the total profits of Rs 6,00,000, the first 2,00,000 are to be distributed in the ratio of 3 : 1. Out of this the share of Y is 1/4th i.e. 200,000 × ¼ = Rs 50,000. For the remaining Rs 400,000, it is to be distributed equally the share of Y will be 400,000 × ½ = Rs 200,000. The total share of Y thus becomes 50,000 + 200,000 = Rs 250,000.
9.	1	Their profit ratio will be A 1500 × 12 B 2000 × 9 C 2250 × 8 Ratio is 1 : 1 : 1. ∴ Each person's share is 900/3 = Rs 300.
10.	3	Profit ratio of A, B and C will be $A\ \frac{1}{4}\times\frac{1}{4}\ B\ \frac{1}{5}\times\frac{1}{2}\ C\ \frac{11}{20}\times 1$ ⇒ ratio is $\frac{1}{16}:\frac{1}{10}:\frac{11}{20}=5:8:44.$ Hence share of A = 5/57 × 3420 = Rs 300, Share of B = 8/57 × 3420 = Rs 480 and Share of C = 44/57 × 3420 = Rs 2640